Theory of Elementary Particles
Lecture 12

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Weinberg-Glashow-Salam model

- Local $SU(2)_L \times U(1)_Y$ symmetry
- fermions: l_L, q_L, e_R, u_R, d_R
- Higgs doublet

$$\Psi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \tag{1}$$

  where $Y(\Psi) = 1$, $Q(\phi^+) = 1$, $Q(\phi^0) = 0$
- Higgs mechanism $SU(2)_L \times U(1)_Y \rightarrow U(1)_{em}$: fermions get masses (through the Yukawa couplings), vector bosons get masses; charged (observed) and neutral (at that time not yet observed) currents.

$$\mathcal{L}_{SU(2)\times U(1)} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4 \tag{2}$$

- Gauge fields
- Fermion fields
- Higgs
- Yukawa terms
- We start from $SU_L(2) \times U_Y(1) \to U_{em}(1)$

Gauge fields sector

$$\mathcal{L}_1 = -\frac{1}{4} F^a_{\mu\nu} F^{a\mu\nu} - \frac{1}{4} G_{\mu\nu} G^{\mu\nu} \tag{3}$$

where

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\varepsilon^{abc} A^b_\mu A^c_\nu, \quad a, b, c = 1, 2, 3 \tag{4}$$

and

$$G_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu \tag{5}$$

Fermion sector

$$\mathcal{L}_2 = \sum_{leptons\,and\,quarks} \bar{\psi}\, i\gamma^\mu D_\mu \psi \tag{6}$$

$$D_\mu \psi = \left(\partial_\mu - ig\,\mathbf{T}\cdot\mathbf{A}_\mu - ig'\frac{Y}{2}B_\mu\right)\psi \tag{7}$$

Let us remind that (Lec. 10)

$$Y(l_L) = -1,\ Y(q_L) = 1/3,\ Y(e_R) = -2,\ Y(u_R) = 4/3,\ Y(d_R) = -2/3 \tag{8}$$

Then for leptons

$$D_\mu l_L = \left(\partial_\mu - i\frac{g}{2}\,\boldsymbol{\tau}\cdot\mathbf{A}_\mu + i\frac{g'}{2}B_\mu\right) l_L, \quad D_\mu e_R = \left(\partial_\mu + ig'B_\mu\right) e_R \tag{9}$$

For quarks

$$D_\mu q_L = \left(\partial_\mu - i\frac{g}{2}\,\boldsymbol{\tau}\cdot\mathbf{A}_\mu - i\frac{g'}{6}B_\mu\right) q_L, \tag{10}$$

$$D_\mu u_R = \left(\partial_\mu - ig'\frac{2}{3}B_\mu\right) u_R, \quad D_\mu d_R = \left(\partial_\mu + i\frac{g'}{3}B_\mu\right) d_R \tag{11}$$

$$
\begin{aligned}
\mathcal{L}_2 \quad = \quad & \overline{l_L}\, i\gamma^\mu \left(\partial_\mu - i\frac{g}{2}\,\boldsymbol{\tau}\cdot\mathbf{A}_\mu + i\frac{g'}{2}B_\mu\right) l_L + \overline{e_R}\, i\gamma^\mu \left(\partial_\mu + ig'B_\mu\right) e_R \\
& + \overline{q_L}\, i\gamma^\mu \left(\partial_\mu - i\frac{g}{2}\,\boldsymbol{\tau}\cdot\mathbf{A}_\mu - i\frac{g'}{6}B_\mu\right) q_L \\
& + \overline{u_R}\, i\gamma^\mu \left(\partial_\mu - ig'\frac{2}{3}B_\mu\right) u_R + \overline{d_R}\, i\gamma^\mu \left(\partial_\mu + i\frac{g'}{3}B_\mu\right) d_R \qquad (12)
\end{aligned}
$$

Higgs sector

$$\mathcal{L}_3 = \left\{ \left(\partial_\mu - \frac{i}{2} g \boldsymbol{\tau} \cdot \mathbf{A}_\mu - \frac{i}{2} g' B_\mu \right) \Psi \right\}^\dagger \left(\partial^\mu - \frac{i}{2} g \boldsymbol{\tau} \cdot \mathbf{A}_\mu - \frac{i}{2} g' B^\mu \right) \Psi - V(\Psi) \tag{13}$$

and

$$V(\Psi) = -\mu^2 \Psi^\dagger \Psi + \lambda (\Psi^\dagger \Psi)^2 \tag{14}$$

Yukawa sector

$$\mathcal{L}_4 = -f^{(e)}\bar{l}_L \Psi e_R - f^{(u)}\overline{q}_L \tilde{\Psi} u_R - f^{(d)}\overline{q}_L \Psi d_R + h.c. \tag{15}$$

and

$$\tilde{\Psi} = i\tau_2 \Psi^* \tag{16}$$

- Yukawa couplings $f^{(e)}$, $f^{(u)}$, and $f^{(d)}$;
- Yukawa contact interactions eHl, qHu, and dHq.

Why $Y(\Psi) = 1$

Notice that

$$\bar{l}_L \Psi e_R \underbrace{\rightarrow}_{U(1)_Y} \bar{l}_L e^{-ig' Y_{l_L}} e^{ig' Y_\Psi} \Psi e^{ig' Y_{e_R}} e_R \Rightarrow \underbrace{-Y_{l_L}}_{+1} + \underbrace{Y_{e_R}}_{-2} + Y_\Psi = 0 \tag{17}$$

so

$$Y(\Psi) = 1$$

while

$$Y(\tilde{\Psi}) = -1$$

Higgs mechanism

$$\langle \Psi \rangle_0 = \langle 0 \mid \Psi \mid 0 \rangle = \begin{pmatrix} 0 \\ \frac{v}{\sqrt{2}} \end{pmatrix} \quad \text{with} \quad v = \sqrt{\frac{\mu^2}{\lambda}} \tag{18}$$

In polar variables

$$\Psi(x) = U^{-1}(\boldsymbol{\xi}) \begin{pmatrix} 0 \\ \frac{v + \eta(x)}{\sqrt{2}} \end{pmatrix} \tag{19}$$

with

$$U(\boldsymbol{\xi}) = \exp\left(i\boldsymbol{\xi} \cdot \boldsymbol{\tau}/v\right) \tag{20}$$

where

$$\langle 0 \mid \boldsymbol{\xi} \mid 0 \rangle = 0, \quad \langle 0 \mid \eta \mid 0 \rangle = 0 \tag{21}$$

We make a gauge transformation:

$$\Psi' = U(\boldsymbol{\xi})\Psi \tag{22}$$

$$\Psi' = \frac{v+\eta(x)}{\sqrt{2}}\chi, \quad \chi = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{23}$$

$$l'_L = U(\boldsymbol{\xi}) l_L, \quad e'_R = e_R \tag{24}$$

$$q'_L = U(\boldsymbol{\xi}) q_L, \quad u'_R = u_R, \quad d'_R = d_R \tag{25}$$

$$\frac{\boldsymbol{\tau}\cdot\mathbf{A}'_\mu}{2} = U(\boldsymbol{\xi})\left(\frac{\boldsymbol{\tau}\cdot\mathbf{A}_\mu}{2}\right)U^{-1}(\boldsymbol{\xi}) - \frac{i}{g}\partial_\mu U(\boldsymbol{\xi})\, U^{-1}(\boldsymbol{\xi}) \tag{26}$$

$$B'_\mu = B_\mu \tag{27}$$

- after redefinition of fields $\mathcal{L}_1$ has the same structure but interactions between $W^{\pm}$, A_μ and Z_0 are present.

Let us consider $\mathcal{L}_3$

$$
\begin{aligned}
\mathcal{L}_3 \quad &\rightarrow \\
*_3 \quad &\equiv \quad \left\{ \left(\partial_\mu - \frac{i}{2} g \boldsymbol{\tau} \cdot \mathbf{A}_\mu - \frac{i}{2} g' \, B'_\mu \right) U^{-1}(\boldsymbol{\xi}) \Psi' \right\}^\dagger \\
&\quad \left(\partial^\mu - \frac{i}{2} g \boldsymbol{\tau} \cdot \mathbf{A}_\mu - \frac{i}{2} g' B'^\mu \right) U^{-1}(\boldsymbol{\xi}) \Psi' \\
&= \quad \Psi'^\dagger \left(\overleftarrow{\partial}_\mu U(\boldsymbol{\xi}) + \partial_\mu U(\boldsymbol{\xi}) - \frac{i}{2} g U(\boldsymbol{\xi}) \boldsymbol{\tau} \cdot \mathbf{A}_\mu - \frac{i}{2} g' \, U(\boldsymbol{\xi}) B'_\mu \right) \qquad (28) \\
&\quad \left(U^{-1}(\boldsymbol{\xi}) \partial^\mu + (\partial^\mu U^{-1}(\boldsymbol{\xi})) + \frac{i}{2} g \boldsymbol{\tau} \cdot \mathbf{A}_\mu U^{-1}(\boldsymbol{\xi}) + \frac{i}{2} g' U^{-1}(\boldsymbol{\xi}) B'^\mu \right) \Psi' \\
&= \quad \Psi'^\dagger \left(\overleftarrow{\partial}_\mu + \partial_\mu U(\boldsymbol{\xi}) U^{-1}(\boldsymbol{\xi}) + \frac{i}{2} g U(\boldsymbol{\xi}) \boldsymbol{\tau} \cdot \mathbf{A}_\mu U^{-1}(\boldsymbol{\xi}) + \frac{i}{2} g' \, B'_\mu \right) \qquad (29) \\
&\quad \left(\partial^\mu - (\partial^\mu U(\boldsymbol{\xi})) U^{-1}(\boldsymbol{\xi}) - \frac{i}{2} g U(\boldsymbol{\xi}) \boldsymbol{\tau} \cdot \mathbf{A}_\mu U^{-1}(\boldsymbol{\xi}) - \frac{i}{2} g' B'^\mu \right) \Psi' \\
&= \quad \Psi'^\dagger \left(\overleftarrow{\partial}_\mu + \frac{i}{2} g \boldsymbol{\tau} \cdot \mathbf{A}'_\mu + \frac{i}{2} g' \, B'_\mu \right) \left(\partial^\mu - \frac{i}{2} g \boldsymbol{\tau} \cdot \mathbf{A}'_\mu - \frac{i}{2} g' B'^\mu \right) \Psi' \\
&= \quad \partial_\mu \Psi'^\dagger \partial^\mu \Psi' - i \partial_\mu \Psi'^\dagger \left(\frac{g}{2} \boldsymbol{\tau} \cdot \mathbf{A}'^\mu + \frac{g'}{2} B'^\mu \right) \Psi' \\
&\quad + i \Psi'^\dagger \left(\frac{g}{2} \boldsymbol{\tau} \cdot \mathbf{A}'^\mu + \frac{g'}{2} B'^\mu \right) \partial_\mu \Psi' \\
&\quad + \Psi'^\dagger \left(\frac{g}{2} \boldsymbol{\tau} \cdot \mathbf{A}'_\mu + \frac{g'}{2} \, B'_\mu \right) \left(\frac{g}{2} \boldsymbol{\tau} \cdot \mathbf{A}'^\mu + \frac{g'}{2} \, B'^\mu \right) \Psi' \qquad (30)
\end{aligned}
$$

$$*_3 = \frac{1}{2}\partial_\mu\eta\partial^\mu\eta + \frac{(v+\eta)^2}{2}\underbrace{\chi^\dagger\left(\frac{g}{2}\boldsymbol{\tau}\cdot\mathbf{A'}_\mu + \frac{g'}{2}B'_\mu\right)\left(\frac{g}{2}\boldsymbol{\tau}\cdot\mathbf{A'}^\mu + \frac{g'}{2}B'^\mu\right)\chi}_{*_{3b}} \quad (31)$$

where

$$*_{3b} = \frac{1}{4}\chi^\dagger\left(g\boldsymbol{\tau}\cdot\mathbf{A'}_\mu + g'\,B'_\mu\right)\left(g\boldsymbol{\tau}\cdot\mathbf{A'}^\mu + g'\,B'^\mu\right)\chi \quad (32)$$

$$= \frac{1}{4}\left(g^2\mathbf{A'}_\mu\cdot\mathbf{A'}^\mu + g'^2B'_\mu B'^\mu - 2gg'A'^3_\mu B'^\mu\right) \quad (33)$$

$$= \frac{1}{4}g^2\left(A'^1_\mu A'^{1\mu} + A'^2_\mu A'^{2\mu}\right) + \frac{1}{4}\left(g'B'_\mu - gA'^3_\mu\right)\left(g'B'^\mu - gA'^{3\mu}\right) \quad (34)$$

We use the property:

$$g\boldsymbol{\tau}\cdot\mathbf{A'}_\mu + g'B'_\mu = \begin{pmatrix} gA'^3_\mu + g'B'_\mu & g(A'^1_\mu - iA'^2_\mu) \\ g(A'^1_\mu + iA'^2_\mu) & -gA'^3_\mu + g'B'_\mu \end{pmatrix}$$

and

$$(g\boldsymbol{\tau}\cdot\mathbf{A'}_\mu + g'B'_\mu)\chi = \begin{pmatrix} g(A'^1_\mu - iA'^2_\mu) \\ -gA'^3_\mu + g'B'_\mu \end{pmatrix}$$

$$\chi^\dagger(g\boldsymbol{\tau}\cdot\mathbf{A'}_\mu + g'B'_\mu) = \begin{pmatrix} g(A'^1_\mu + iA'^2_\mu) & -gA'^3_\mu + g'B'_\mu \end{pmatrix}$$

Let us introduce

$$W_\mu^\pm = \frac{1}{\sqrt{2}}\left(A'^1_\mu \mp iA'^2_\mu\right) \tag{35}$$

then

$$*_{3b} \quad = \quad \frac{g^2}{2}W_\mu^+ W^{-\mu} + \frac{1}{4}\left(g'B'_\mu - gA'^3_\mu\right)\left(g'B'^\mu - gA'^{3\mu}\right) \tag{36}$$

We obtained the mass term for $W^{\pm}$, which reads

$$\frac{v^2 g^2}{4} W_{\mu}^{+} W^{-\mu} = M_W^2 W_{\mu}^{+} W^{-\mu} \tag{37}$$

so

$$M_W^2 = \frac{g^2 v^2}{4} \tag{38}$$

Consider the mass terms

$$*_{3bZ} = \frac{v^2}{8}\left(g' B'_\mu - g A'^3_\mu\right)\left(g' B'^\mu - g A'^{3\mu}\right) \quad (39)$$

$$= \frac{v^2}{8}\begin{pmatrix} A'^3_\mu & B'_\mu \end{pmatrix}\begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix}\begin{pmatrix} A'^{3\mu} \\ B'^\mu \end{pmatrix} \quad (40)$$

Notice that

$$\det\begin{pmatrix} g^2-\lambda & -gg' \\ -gg' & g'^2-\lambda \end{pmatrix} = \lambda(\lambda - g^2 - g'^2) \quad (41)$$

The eigenvalues are 0 and $g^2 + g'^2$

$$*_{3bZ} = \frac{1}{2}\begin{pmatrix} Z_\mu & A_\mu \end{pmatrix}\begin{pmatrix} M_Z^2 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} Z^\mu \\ A^\mu \end{pmatrix} \quad (42)$$

where

$$M_Z^2 = v^2(g^2 + g'^2)/4 \quad (43)$$

$$Z_\mu = \cos\theta_W A'^3_\mu - \sin\theta_W B'_\mu \quad (44)$$

$$A_\mu = \sin\theta_W A'^3_\mu + \cos\theta_W B'_\mu \quad (45)$$

We easily get

$$*_{3bZ} \quad = \quad \frac{1}{2} M_Z^2 Z_\mu Z^\mu \tag{46}$$

so

$$M_Z^2 (\cos\theta_W A'^3_\mu - \sin\theta_W B'_\mu)^2 \quad = \quad \frac{v^2}{4} (g A'^3_\mu - g' B'_\mu)^2 \tag{47}$$

$$(g^2 + g'^2)(\cos\theta_W A'^3_\mu - \sin\theta_W B'_\mu)^2 \quad = \quad (g A'^3_\mu - g' B'_\mu)^2 \tag{48}$$

so

$$\cos\theta_W = \frac{g}{\sqrt{g^2 + g'^2}}, \quad \sin\theta_W = \frac{g'}{\sqrt{g^2 + g'^2}} \tag{49}$$

hence

$$\tan\theta_W = \frac{g'}{g} \tag{50}$$

θ_W is the Weinberg angle. Notice that

$$\rho \equiv \frac{M_W^2}{M_Z^2 \cos^2\theta_W} = 1 \tag{51}$$

The potential term reads

$$V(\Psi') = -\mu^2 \frac{(v+\eta(x))^2}{2} + \lambda \frac{(v+\eta(x))^4}{4} \tag{52}$$

$$= -\frac{\mu^2}{2}\left(v^2+\eta^2(x)+2v\eta(x)\right)$$

$$+\frac{\lambda}{4}\left(v^4+\eta^4(x)+4v^3\eta(x)+6v^2\eta^2(x)+4v\eta^3(x)\right) \tag{53}$$

$$= -\frac{\mu^2 v^2}{4} + \mu^2\eta^2(x) + \frac{\mu^2}{v}\eta^3(x) + \frac{\mu^2}{4v^2}\eta^4(x) \tag{54}$$

Notice that the η Higgs particle (neutral) gets the mass $m_\eta = \sqrt{2}\mu$

$$\mathcal{L}_3 = \frac{1}{2}\partial_\mu\eta\partial^\mu\eta + M_W^2 W_\mu^+ W^{-\mu} + \frac{M_W^2}{v^2}\eta^2 W_\mu^+ W^{-\mu} + 2M_W^2\frac{\eta}{v}W_\mu^+ W^{-\mu}$$
$$+\frac{M_Z^2}{2}Z_\mu Z^\mu + \eta^2\frac{M_Z^2}{2v^2}Z_\mu Z^\mu + \frac{M_Z^2}{v}\eta Z_\mu Z^\mu - V(\Psi') \quad (55)$$
$$= \frac{1}{2}\partial_\mu\eta\partial^\mu\eta - \mu^2\eta^2(x) \quad (56)$$
$$+M_W^2 W_\mu^+ W^{-\mu} + \frac{M_Z^2}{2}Z_\mu Z^\mu \quad (57)$$
$$+\frac{g^2}{4}\eta^2 W_\mu^+ W^{-\mu} + gM_W\,\eta W_\mu^+ W^{-\mu} + \frac{g^2}{8\cos^2\theta_W}\eta^2 Z_\mu Z^\mu + \frac{gM_Z}{2\cos\theta_W}\eta Z_\mu Z^\mu \quad (58)$$
$$+\frac{\mu^2 M_W^2}{g^2} - \frac{\mu^2 g}{2M_W}\eta^3(x) - \frac{\mu^2 g^2}{16M_W^2}\eta^4(x) \quad (59)$$

- Higgs field, $W\pm$ and Z^0 mass terms, the Higgs coupling with gauge weak bosons.

Now consider $\mathcal{L}_4$

$$\mathcal{L}_4 = -f^{(e)}\overline{l'}_L \Psi' e'_R - f^{(u)}\overline{q'}_L \tilde{\Psi}' u'_R - f^{(d)}\overline{q'}_L \Psi' d'_R + h.c. \quad (60)$$

$$= -\frac{(v+\eta)}{\sqrt{2}}\left(f^{(e)}\overline{e'_L}e'_R + f^{(u)}\overline{u'_L}u'_R + f^{(d)}\overline{d'_L}d'_R\right) + h.c. \quad (61)$$

$$= -\frac{v}{\sqrt{2}}\left(f^{(e)}\overline{e'_L}e'_R + f^{(u)}\overline{u'_L}u'_R + f^{(d)}\overline{d'_L}d'_R\right) \quad (62)$$

$$-\frac{\eta}{\sqrt{2}}\left(f^{(e)}\overline{e'_L}e'_R + f^{(u)}\overline{u'_L}u'_R + f^{(d)}\overline{d'_L}d'_R\right) + h.c. \quad (63)$$

$$= -\frac{v}{\sqrt{2}}\left(f^{(e)}\overline{e'}e' + f^{(u)}\overline{u'}u' + f^{(d)}\overline{d'}d'\right) - \frac{\eta}{\sqrt{2}}\left(f^{(e)}\overline{e'}e' + f^{(u)}\overline{u'}u' + f^{(d)}\overline{d'}d'\right) \quad (64)$$

We see that electron and quarks get masses:

$$m_e = \frac{vf^{(e)}}{\sqrt{2}} = \sqrt{2}\frac{M_W}{g}f^{(e)}, \quad m_d = \frac{vf^{(d)}}{\sqrt{2}} = \sqrt{2}\frac{M_W}{g}f^{(d)}, \quad m_u = \frac{vf^{(u)}}{\sqrt{2}} = \sqrt{2}\frac{M_W}{g}f^{(u)} \quad (65)$$

Now consider $\mathcal{L}_2$

$$
\begin{aligned}
\mathcal{L}_2 &= \overline{l'_L} i\gamma^\mu \left(\partial_\mu - i\frac{g}{2}\boldsymbol{\tau}\cdot\mathbf{A'}_\mu + i\frac{g'}{2}B'_\mu\right) l'_L + \overline{e'_R} i\gamma^\mu \left(\partial_\mu + ig'B'_\mu\right) e'_R \\
&+ \overline{q'_L} i\gamma^\mu \left(\partial_\mu - i\frac{g}{2}\boldsymbol{\tau}\cdot\mathbf{A'}_\mu - i\frac{g'}{6}B'_\mu\right) q'_L \qquad (66) \\
&\quad + \overline{u'_R} i\gamma^\mu \left(\partial_\mu - ig'\frac{2}{3}B'_\mu\right) u'_R + \overline{d'_R} i\gamma^\mu \left(\partial_\mu + i\frac{g'}{3}B'_\mu\right) d'_R \\
&= \overline{l'_L} i\gamma^\mu \partial_\mu l'_L + \overline{e'_R} i\gamma^\mu \partial_\mu e'_R + \overline{q'_L} i\gamma^\mu \partial_\mu q'_L + \overline{u'_R} i\gamma^\mu \partial_\mu u'_R + \overline{d'_R} i\gamma^\mu \partial_\mu d'_R \\
&\quad \frac{g}{2}\mathbf{A'}_\mu \cdot \left(\overline{l'_L}\gamma^\mu \boldsymbol{\tau} l'_L + \overline{q'_L}\gamma^\mu \boldsymbol{\tau} q'_L\right) \\
&\quad - \frac{g'}{2}B'_\mu \left(\overline{l'_L}\gamma^\mu l'_L + 2\overline{e'_R}\gamma^\mu e'_R - \frac{1}{3}\overline{q'_L}\gamma^\mu q'_L - \frac{4}{3}\overline{u'_R}\gamma^\mu u'_R + \frac{2}{3}\overline{d'_R}\gamma^\mu d'_R\right) \qquad (67)
\end{aligned}
$$

Notice that

$$\tau_\pm = \frac{\tau_1 \pm i\tau_2}{2} \Rightarrow \tau_1 = \tau_+ + \tau_-, \quad \tau_2 = i(\tau_- - \tau_+) \tag{68}$$

and

$$A_1 = \frac{1}{\sqrt{2}}(W^+ + W^-) \quad A_2 = \frac{i}{\sqrt{2}}(W^+ - W^-) \tag{69}$$

$$\begin{aligned} A'_1\tau_1 + A'_2\tau_2 &= \frac{1}{\sqrt{2}}\left(W^+\tau_+ + W^+\tau_- + W^-\tau_+ + W^-\tau_-\right) && (70)\\ &\quad -\frac{1}{\sqrt{2}}\left(-W^+\tau_+ + W^+\tau_- + W^-\tau_+ - W^-\tau_-\right) && (71)\\ &= \sqrt{2}\left(W^+\tau^+ + W^-\tau^-\right) && (72) \end{aligned}$$

Hence

$$\begin{aligned} * &= \frac{g}{2}\mathbf{A}'_\mu \cdot \left(\overline{l'_L}\gamma^\mu \boldsymbol{\tau} l'_L + \overline{q'_L}\gamma^\mu \boldsymbol{\tau} q'_L\right) && (73)\\ &\rightarrow \frac{g}{\sqrt{2}}W^+_\mu\left(\overline{\nu'_L}\gamma^\mu e'_L + \overline{u'_L}\gamma^\mu d'_L\right) + \frac{g}{\sqrt{2}}W^-_\mu\left(\overline{e'_L}\gamma^\mu \nu'_L + \overline{d'_L}\gamma^\mu u'_L\right) + ... && (74) \end{aligned}$$

We define charged current:

$$J_\mu^+ = \overline{l'_L}\gamma_\mu\tau_+ l'_L + \overline{q'_L}\gamma_\mu\tau_+ q'_L = \frac{1}{2}\left(\overline{\nu'}\gamma^\mu(1-\gamma_5)e' + \overline{u'}\gamma^\mu(1-\gamma_5)d'\right) \qquad (75)$$

Now consider the neutral current part:

$$A'^{3}_{\mu} = \cos\theta_W Z_\mu + \sin\theta_W A_\mu \quad (76)$$

$$B'_{\mu} = -\sin\theta_W Z_\mu + \cos\theta_W A_\mu \quad (77)$$

then

$$\begin{aligned} ** &= \frac{g}{2} A'^{3}_{\mu} \left(\overline{l'_L}\gamma^\mu \tau_3 l'_L + \overline{q'_L}\gamma^\mu \tau_3 q'_L\right) \\ &\quad -\frac{g'}{2} B'_\mu \left(\overline{l'_L}\gamma^\mu l'_L + 2\overline{e'_R}\gamma^\mu e'_R - \frac{1}{3}\overline{q'_L}\gamma^\mu q'_L - \frac{4}{3}\overline{u'_R}\gamma^\mu u'_R + \frac{2}{3}\overline{d'_R}\gamma^\mu d'_R\right) \quad (78) \\ &= g A'^{3}_{\mu} J^\mu_3 + \frac{g'}{2} B'_\mu J^\mu_Y \quad (79) \end{aligned}$$

Notice that the hyper-charge current reads

$$
\begin{aligned}
J^{\mu}_{Y} &= -\overline{l'_L}\gamma^{\mu} l'_L - 2\overline{e'_R}\gamma^{\mu} e'_R + \frac{1}{3}\overline{q'_L}\gamma^{\mu} q'_L + \frac{4}{3}\overline{u'_R}\gamma^{\mu} u'_R - \frac{2}{3}\overline{d'_R}\gamma^{\mu} d'_R && (80)\\
&= -\overline{\nu'_L}\gamma^{\mu}\nu'_L - \overline{e'_L}\gamma^{\mu} e'_L - 2\overline{e'_R}\gamma^{\mu} e'_R + \frac{1}{3}\overline{u'_L}\gamma^{\mu} u'_L + \frac{1}{3}\overline{d'_L}\gamma^{\mu} d'_L \\
&\quad + \frac{4}{3}\overline{u'_R}\gamma^{\mu} u'_R - \frac{2}{3}\overline{d'_R}\gamma^{\mu} d'_R && (81)\\
&= 2\left(-\overline{e'}\gamma^{\mu} e' + \frac{2}{3}\overline{u'}\gamma^{\mu} u' - \frac{1}{3}\overline{d'}\gamma^{\mu} d'\right) && (82)\\
&\quad -2\cdot\frac{1}{2}\cdot\left(\overline{\nu'_L}\gamma^{\mu}\nu'_L - \overline{e'_L}\gamma^{\mu} e'_L + \overline{u'_L}\gamma^{\mu} u'_L - \overline{d'_L}\gamma^{\mu} d'_L\right) \\
&= 2(J^{\mu}_{em} - J^{\mu}_{3}) && (83)
\end{aligned}
$$

where

$$
\begin{aligned}
J^{\mu}_{em} &= -\overline{e'_L}\gamma^{\mu} e'_L - \overline{e'_R}\gamma^{\mu} e'_R + \frac{2}{3}\overline{u'_L}\gamma^{\mu} u'_L + \frac{2}{3}\overline{u'_R}\gamma^{\mu} u'_R - \frac{1}{3}\overline{d'_L}\gamma^{\mu} d'_L - \frac{1}{3}\overline{d'_R}\gamma^{\mu} d'_R \\
&= -\overline{e'}\gamma^{\mu} e' + \frac{2}{3}\overline{u'}\gamma^{\mu} u' - \frac{1}{3}\overline{d'}\gamma^{\mu} d' && (84)
\end{aligned}
$$

Now consider the neutral current part:

$$A'^{3}_{\mu} = \cos\theta_W Z_\mu + \sin\theta_W A_\mu \quad (85)$$

$$B'_{\mu} = -\sin\theta_W Z_\mu + \cos\theta_W A_\mu \quad (86)$$

then

$$\begin{aligned} ** &= g(\cos\theta_W Z_\mu + \sin\theta_W A_\mu) J_3^\mu + g'(\cos\theta_W A_\mu - \sin\theta_W Z_\mu)\left(J_{em}^\mu - J_3^\mu\right) \\ &= g'\cos\theta_W A_\mu J_{em}^\mu + A_\mu J_3^\mu (g\sin\theta_W - g'\cos\theta_W) \quad (87) \\ &\quad + Z_\mu\left(g\cos\theta_W J_3^\mu - g'\sin\theta_W\left(J_{em}^\mu - J_3^\mu\right)\right) \quad (88) \\ &= g\sin\theta_W A_\mu J_{em}^\mu + \frac{g}{\cos\theta_W} Z_\mu\left(J_3^\mu - \sin^2\theta_W J_{em}^\mu\right) \quad (89) \end{aligned}$$

Let

$$J_0^\mu \equiv J_3^\mu - \sin^2\theta_W J_{em}^\mu \quad (90)$$

hence

$$e = g\sin\theta_W \quad (91)$$

Notice that

$$J_0^\mu = \sum_f \left(g_L^f \overline{f_L} \gamma^\mu f_L + g_R^f \overline{f_R} \gamma^\mu f_R \right) \tag{92}$$

where

$$g_{L,R}^f = T_3(f_{L,R}) - Q(f) \sin^2 \theta_W \tag{93}$$

hence

$$g_L^\nu = \frac{1}{2} \qquad g_R^\nu = 0 \tag{94}$$

$$g_L^e = -\frac{1}{2} + \sin^2 \theta_W \qquad g_R^e = \sin^2 \theta_W \tag{95}$$

$$g_L^u = \frac{1}{2} - \frac{2}{3} \sin^2 \theta_W \qquad g_R^u = -\frac{2}{3} \sin^2 \theta_W \tag{96}$$

$$g_L^d = -\frac{1}{2} + \frac{1}{3} \sin^2 \theta_W \qquad g_R^d = \frac{1}{3} \sin^2 \theta_W \tag{97}$$

$$\mathcal{L}_2 = \underbrace{\overline{l'_L} i\gamma^\mu \partial_\mu l'_L + \overline{e'_R} i\gamma^\mu \partial_\mu e'_R + \overline{q'_L} i\gamma^\mu \partial_\mu q'_L + \overline{u'_R} i\gamma^\mu \partial_\mu u'_R + \overline{d'_R} i\gamma^\mu \partial_\mu d'_R}_{kinetic\,terms}$$

$$+\frac{g}{2\sqrt{2}} W^+_\mu \left(\overline{\nu'}\gamma^\mu(1-\gamma_5)e' + \overline{u'}\gamma^\mu(1-\gamma_5)d'\right) + \frac{g}{2\sqrt{2}} W^-_\mu \left(\overline{e'}\gamma^\mu(1-\gamma_5)\nu' + \overline{d'}\gamma^\mu$$

$$+eA_\mu \left(-\overline{e'}\gamma^\mu e' + \frac{2}{3}\overline{u'}\gamma^\mu u' - \frac{1}{3}\overline{d'}\gamma^\mu d'\right)$$

$$+\frac{g}{\cos\theta_W} Z_\mu \sum_f \left(g^f_L \overline{f_L}\gamma^\mu f_L + g^f_R \overline{f_R}\gamma^\mu f_R\right)$$

- Charged Current (CC) interactions, ElectroMagnetic (EM) interactions, Neutral Current (NC) interactions

The minimal model for electroweak interactions, only for one fermion family, contains seven arbitrary parameters

$$e, \quad \sin\theta_W, \quad M_W, \quad m_\eta, \quad m_e, \quad m_u, \quad m_d \tag{102}$$

$$\mathcal{L}_1 = -\frac{1}{4} F'^{a}_{\mu\nu} F'^{a\mu\nu} - \frac{1}{4} G'_{\mu\nu} G'^{\mu\nu} \tag{103}$$

$$\begin{aligned}\mathcal{L}_2 = & \overline{l'_L} i\gamma^\mu \partial_\mu l'_L + \overline{e'_R} i\gamma^\mu \partial_\mu e'_R + \overline{q'_L} i\gamma^\mu \partial_\mu q'_L + \overline{u'_R} i\gamma^\mu \partial_\mu u'_R + \overline{d'_R} i\gamma^\mu \partial_\mu d'_R \\ & + \frac{g}{2\sqrt{2}} \left[W^+_\mu \left(\overline{\nu'} \gamma^\mu (1-\gamma_5) e' + \overline{u'} \gamma^\mu (1-\gamma_5) d' \right) + W^-_\mu \left(\overline{e'} \gamma^\mu (1-\gamma_5) \nu' + \overline{d'} \gamma^\mu (1-\gamma_5) u' \right) \right] \\ & + e A_\mu \left(-\overline{e'} \gamma^\mu e' + \frac{2}{3} \overline{u'} \gamma^\mu u' - \frac{1}{3} \overline{d'} \gamma^\mu d' \right) + \frac{g}{\cos\theta_W} Z_\mu \sum_f \left(g^f_L \overline{f_L} \gamma^\mu f_L + g^f_R \overline{f_R} \gamma^\mu f_R \right)\end{aligned} \tag{104}$$

$$\begin{aligned}\mathcal{L}_3 = & \frac{1}{2} \partial_\mu \eta \partial^\mu \eta - \mu^2 \eta^2(x) \\ & + M_W^2 W^+_\mu W^{-\mu} + \frac{M_Z^2}{2} Z_\mu Z^\mu \\ & + \frac{g^2}{4} \eta^2 W^+_\mu W^{-\mu} + g M_W \eta W^+_\mu W^{-\mu} + \frac{g^2}{8\cos^2\theta_W} \eta^2 Z_\mu Z^\mu + \frac{g M_Z}{2\cos\theta_W} \eta Z_\mu Z^\mu \\ & + \frac{\mu^2 M_W^2}{g^2} - \frac{\mu^2 g}{2 M_W} \eta^3(x) - \frac{\mu^2 g^2}{16 M_W^2} \eta^4(x)\end{aligned} \tag{105}$$

$$\mathcal{L}_4 = -m_e \overline{e'} e' - m_u \overline{u'} u' - m_d \overline{d'} d' - \frac{g}{2 M_W} \left(m_e \overline{e'} e' + m_u \overline{u'} u' + m_d \overline{d'} d' \right) \eta \tag{106}$$

The lagrangian in Unitary gauge!

- Consider $e + \nu_e \to e + \nu_e$ charged current process

$$\mathcal{L}_{int}(x) = -\frac{G_F}{\sqrt{2}} \mathcal{J}^{\alpha}(x) \mathcal{J}^{\dagger}_{\alpha}(x), \tag{107}$$

where

$$\mathcal{J}_{\mu}(x) = \overline{e}(x) \gamma_{\mu} \left(1 - \gamma_5\right) \nu_e(x) \tag{108}$$

Notice that

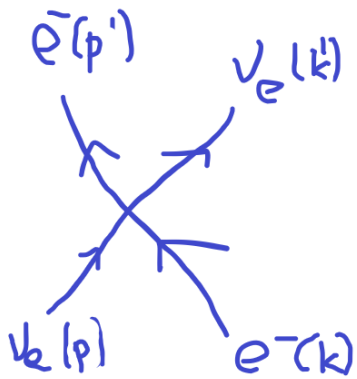
$$\mathcal{H}_{int}(x) = -\mathcal{L}_{int}(x) \tag{109}$$

In the Fermi model the effective vertex reads

$$-i\frac{G_F}{\sqrt{2}} \tag{110}$$

hence the scattering amplitude reads

$$i\mathcal{M} = -i\frac{G_F}{\sqrt{2}}\,\bar{\nu}(k')\gamma_\mu(1-\gamma_5)e(k)\;\bar{e}(p')\gamma^\mu(1-\gamma_5)\nu(p) \tag{111}$$



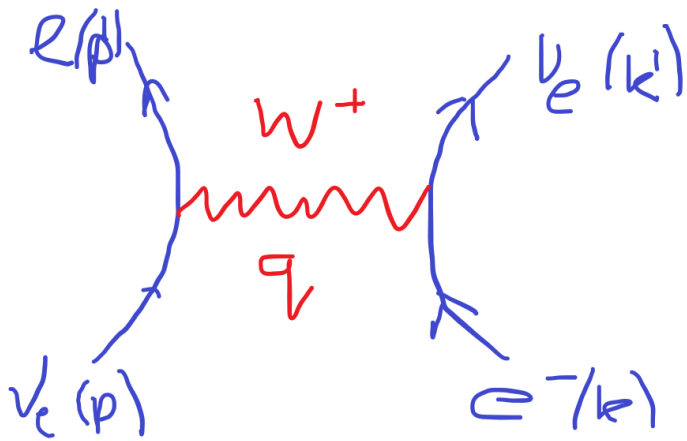
- In the $SU(2) \times U(1)$ model we get

$$i\mathcal{M} = i\frac{1}{2}\frac{g}{\sqrt{2}}\bar{\nu}(k')\gamma_\mu(1-\gamma_5)e(k)\left[-i\frac{g^{\mu\nu}-q_\mu q_\nu/M_W^2}{q^2-M_W^2}\right]\bar{e}(p')\gamma_\nu(1-\gamma_5)\nu(p)i\frac{1}{2}\frac{g}{\sqrt{2}}, \tag{112}$$

  where $q = k - k'$.

- Note that the weak νWe vertex reads:

$$\frac{ig}{2\sqrt{2}}\gamma_\mu(1-\gamma_5) \tag{113}$$



If $q^2 \ll M_W^2$ and $q_\mu \ll M_W$ then

$$\left[-i\frac{g^{\mu\nu} - q_\mu q_\nu / M_W^2}{q^2 - M_W^2}\right] \to -\frac{1}{M_W^2}\left[-i\frac{g^{\mu\nu} - q_\mu q_\nu / M_W^2}{1 - q^2/M_W^2}\right] \approx \frac{i}{M_W^2}\left[g^{\mu\nu}\right] \tag{114}$$

$$(112) \to -i\frac{g^2}{8}\frac{1}{M_W^2}\bar{\nu}(k')\gamma^\mu(1-\gamma_5)e(k)\ \bar{e}(p')\gamma_\mu(1-\gamma_5)\nu(p) \tag{115}$$

then we get equality

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}. \tag{116}$$

so

$$g^2 = 4\sqrt{2}G_F M_W^2 \tag{117}$$

Adler, Bell and Jackiw (ABJ) anomaly

- the validity of the Ward identity is not automatic when there are fermions in the theory!
- appearing anomalies break the renormalizability of the theory.
- the anomalies are cancelled within each fermion family, hence we need to always consider one lepton and one quark family together with the color degrees of freedom.